Flight Dynamics & Control

Tutorial 3

1. The longitudinal quartic for a mid-sized transport aircraft is

$$\lambda^4 + 2.00\lambda^3 + 3.01\lambda^2 + x\lambda + 0.03 = 0.$$

   (a) If the aircraft's phugoid mode is neutrally (un)stable, find the value of x, which is small.

   (b) Hence find the period and damping of all the longitudinal natural modes of the aircraft.

   (c) Based on your findings, what is the cruising speed of the aircraft?

(a) if phugoid mode is neutrally stable, then $\lambda = 0 \pm ai = \pm ai$

$\therefore (\pm ai)^4 + 0.03(\pm ai)^3 + 3.01(\pm ai)^2 \pm xai + 0.03 = 0$

$\therefore a^4 \mp 2a^3 i - 3.01a^2 \pm xai + 0.03 = 0$

$\therefore (a^4 - 3.01a^2 + 0.03)[\pm(-2a^3 + xa)i] = 0$

for $a^4 - 3.01a^2 + 0.03 = 0$, $a^2 = \dfrac{3.01 \pm \sqrt{8.9401}}{2} = 3$ or 0.01

$\therefore a^2 = 0.01$ for phugoid mode which is relatively small

$\therefore a = \pm 0.1$

for $2a^3 - xa = 0$, $0.002 - 0.1x = 0$

$\therefore x = 0.02$

(a) By Routh's discriminant, $R = 0$ for neutral stable

$$R = BCD - B^2E - AD^2 = 6.02x - 0.12 - x^2 = 0$$

$$\therefore\ x = \frac{-6.02 \pm \sqrt{35.7604}}{-2} = 6 \text{ or } 0.02$$

$\therefore\ x = 0.02$ for phugoid mode $\quad \therefore\ a = \pm 0.1i$

(b) for phugoid mode, $a = \pm 0.1$ $\quad \therefore\ \lambda = \pm 0.1i$ $\quad \therefore\ \omega_d = \omega_n = 0.1$

$$\therefore\ T = \frac{2\pi}{\omega_d} = \frac{2\pi}{0.1} = 20\pi$$

$$\therefore\ \zeta = \frac{|r_j|}{|\omega_n|} = \frac{0}{0.1} = 0$$

for SPPO mode, $(A\lambda^2 + B\lambda + C)(\lambda - 0.1i)(\lambda + 0.1i) = 0$

$$\therefore\ (A\lambda^2 + B\lambda + C)(\lambda^2 + 0.01) = 0$$

$$\therefore\ A\lambda^4 + B\lambda^3 + (C + 0.01A)\lambda^2 + 0.01B\lambda + 0.01C = 0$$

$\therefore\ A = 1,\ B = 2,\ C = 3$

$\therefore$ for $A\lambda^2 + B\lambda + C$, $\lambda = \dfrac{-2 \pm \sqrt{4 - 12}}{2} = -1 \pm \sqrt{2}i$

$\therefore\ \omega_d = \sqrt{2}$ $\quad \therefore\ T = \dfrac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$

$$\therefore\ \zeta = \frac{|r_j|}{|\omega_n|} = \frac{1}{\sqrt{1^2 + \sqrt{2}^2}} = \frac{1}{3}$$

(c) for phugoid mode, $\omega_d = \sqrt{2}\,\dfrac{g}{U_e} = 0.1$

$$\therefore U_e = \frac{\sqrt{2}\,g}{0.1} = 138.73\ m/s$$

2. The lateral stability quintic of a large airliner is:

$$\lambda^5 + a\lambda^4 + 1.1025\lambda^3 + 1.0025\lambda^2 = 0.$$

The aircraft is known to have a stable roll subsidence mode with a time to half amplitude of 0.6931 s.

(a) Approximate the value of the unknown parameter a.

(b) Will this aircraft be dynamically stable in the spiral mode? What are the physical implications of your findings?

(c) What is the frequency of oscillation and damping ratio of this aircraft's Dutch Roll mode.

(d) What behaviour does the lateral eigenvalue $\lambda = 0$ that aircraft typically exhibit correspond to?

(a) $t_{\frac{1}{2}} = \frac{\ln 2}{|r_j|}$ $\therefore \lambda = r_j$ for roll subsidence $= -1$

$\therefore (-1)^5 + a(-1)^4 + 1.1025(-1)^3 + 1.0025(-1)^2 = 0$

$\therefore -1 + a - 1.1025 + 1.0025 = a - 1.1 = 0$

$\therefore a = 1.1$

(b) $\lambda^5 + 1.1\lambda^4 + 1.1025\lambda^3 + 1.0025\lambda^2 = 0$

$\lambda^2(\lambda^3 + 1.1\lambda^2 + 1.1025\lambda + 1.0025) = 0$

$\therefore$ There are two neutrally stable mode

$\therefore$ Spiral mode would be neutrally stable

$\therefore$ Aircraft will turn at constant angle once input a turn

(c) $\lambda^3 + 1.1\lambda^2 + 1.1025\lambda + 1.0025 = (\lambda+1)(\lambda^2 + B\lambda + C)$

$= \lambda^3 + (B+1)\lambda^2 + (B+C)\lambda + C = 0$

$\therefore B = 0.1,\ C = 1.0025$

$\therefore$ for Dutch roll, $\lambda^2 + 0.1\lambda + 1.0025 = 0$

$$\therefore \lambda = \frac{-0.1 \pm \sqrt{-4}}{2} = -0.05 \pm i$$

$$\therefore \omega_d = 1 \qquad \therefore T = \frac{2\pi}{\omega_d} = 2\pi$$

$$\therefore \xi = \frac{0.05}{\sqrt{0.05^2 + 1^2}} = 0.049938$$

(d) Aircraft is neutrally stable with respect to a change in the defination of its heading angle

3. The dynamics of a rotorcraft can be represented in state space by the system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

$$\begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \begin{bmatrix} -0.4 & 0 & -0.01 \\ 1 & 0 & 0 \\ -1.4 & 9.8 & -0.02 \end{bmatrix} \begin{bmatrix} q \\ \theta \\ U \end{bmatrix} + \begin{bmatrix} 6.3 \\ 0 \\ 9.8 \end{bmatrix} u(t)$$

where $u(t)$ is the control input being applied to the system.

(a) Is the open-loop system stable?

(b) Derive the transfer functions, in frequency domain, for the states of the system with respect to the control input.

(c) Hence calculate the effect of a step control input on the rotorcraft's velocity U

(d) If a feedback control system is implemented such that the control input

$$u(t) = -\begin{bmatrix} 0.4706 & 1 & 0.0627 \end{bmatrix} \mathbf{x}(t),$$

show that the system is reduced to the following representation. What type of controller is that?

$$\begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \begin{bmatrix} -3.3648 & -6.3000 & -0.4050 \\ 1 & 0 & 0 \\ -6.0119 & 0 & -0.6345 \end{bmatrix} \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

(e) For the closed-loop system given in (d), calculate the closed-loop response of the system from an initial condition [1, 1, 1].

(a) for open loop system,

$$\begin{vmatrix} -0.4-\lambda & 0 & -0.01 \\ 1 & -\lambda & 0 \\ -1.4 & 9.8 & -0.02-\lambda \end{vmatrix}$$

$$\therefore \; -1\cdot 9.8\cdot 0.01 - \lambda\left[(-0.4-\lambda)(-0.02-\lambda) - (-0.01)(-1.4)\right] = 0$$

$$\therefore \; -\lambda^3 - 0.42\lambda^2 + 0.006\lambda - 0.098 = 0$$

$\therefore \; \lambda_1 = -0.65662, \quad \lambda_{2,3} = 0.1178 \pm 0.3682i$ $\qquad\qquad \therefore$ unstable

OR

for Routh's Discriminant. $A\lambda^3 + B\lambda^2 + C\lambda + D = 0$

$A, B, C, D > 0$ and $AD > BC$ for stable

(b) $sX(s) = AX(s) + BU(s)$

$\therefore (Is - A)X(s) = BU(s)$

$$\therefore G(s) = \frac{X(s)}{U(s)} = \frac{B}{Is - A} = (Is - A)^{-1}B$$

$$Is - A = \begin{bmatrix} s+0.4 & 0 & 0.01 \\ -1 & s & 0 \\ 1.4 & -9.8 & s+0.02 \end{bmatrix}$$

$$\therefore (Is - A)^{-1} = \frac{adj(Is-A)}{det(Is-A)} = \frac{C^T}{det(Is-A)}$$

$$= \frac{\begin{bmatrix} s^2+0.02s & s+0.02 & -1.4s+9.8 \\ -0.098 & s^2+0.42s-0.006 & 9.8s+3.92 \\ -0.01s & -0.01 & s^2+0.4s \end{bmatrix}^T}{s^3 + 0.42s^2 - 0.006s + 0.098}$$

$$= \frac{\begin{bmatrix} s^2+0.02s & -0.098 & -0.01s \\ s+0.02 & s^2+0.42s-0.006 & -0.01 \\ -1.4s+9.8 & 9.8s+3.92 & s^2+0.4s \end{bmatrix}}{s^3 + 0.42s^2 - 0.006s + 0.098}$$

$$\therefore G(s) = \frac{\begin{bmatrix} s^2+0.02s & -0.098 & -0.01s \\ s+0.02 & s^2+0.42s-0.006 & -0.01 \\ -1.4s+9.8 & 9.8s+3.92 & s^2+0.4s \end{bmatrix}\begin{bmatrix} 6.3 \\ 0 \\ 9.8 \end{bmatrix}}{s^3+0.42s^2-0.006s+0.098}$$

$$= \frac{\begin{bmatrix} 6.3s^2+0.028s \\ 6.3s+0.028 \\ 9.8s^2-4.9s+61.74 \end{bmatrix}}{s^3+0.42s^2-0.006s+0.098}$$

(c) $G(s)$ for $U = \dfrac{9.8s^2-4.9s+61.74}{s^3+0.42s^2-0.006s+0.098} = \dfrac{U(s)}{u(s)}$

where $u(s) = \frac{1}{s} u^*$

$$\therefore \frac{U(s)}{u^*} = \frac{9.8s^2-4.9s+61.74}{s(s+0.65662)(s^2-0.2366s+0.1493)}$$

... see solutions

(d) P controller. where $u = Cx$

$\therefore e = r - u = -Cx$

$\therefore \dot{x} = Ax + Bu = (A - BC)x$

$$\therefore \begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \left(\begin{bmatrix} -0.4 & 0 & -0.01 \\ 1 & 0 & 0 \\ -1.4 & 9.8 & -0.02 \end{bmatrix} - \begin{bmatrix} 6.3 \\ 0 \\ 9.8 \end{bmatrix} [0.4706 \ \ 1 \ \ 0.0627] \right) \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

$$\therefore \begin{bmatrix} \dot{q} \\ \dot{\theta} \\ \dot{U} \end{bmatrix} = \left(\begin{bmatrix} -0.4 & 0 & -0.01 \\ 1 & 0 & 0 \\ -1.4 & 9.8 & -0.02 \end{bmatrix} - \begin{bmatrix} 2.96478 & 6.3 & 0.39501 \\ 0 & 0 & 0 \\ 4.61188 & 9.8 & 0.61446 \end{bmatrix} \right) \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

$$= \begin{bmatrix} -3.36478 & -6.3 & -0.40501 \\ 1 & 0 & 0 \\ -6.01188 & 0 & -0.63446 \end{bmatrix} \begin{bmatrix} q \\ \theta \\ U \end{bmatrix}$$

(e) detail see solution

4. The longitudinal motion of an aerial vehicle is given by

$$\dot{w} = -2w + 197\dot{\theta} - 27\delta$$
$$\ddot{\theta} = -0.25w - 15\dot{\theta} - 45\delta$$

(a) Pick an appropriate set of states and express the dynamics of the system in state-space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}.$$

(b) Determine the stability of the open-loop system.

(c) Determine the proportional state feedback control law

$$\mathbf{u} = \mathbf{K}\mathbf{x}$$

required such that the closed-loop system response is oscillatory with a damping ration $\zeta = 0.5$ and an undamped natural frequency $\omega_n = 20$ rad/s.

(a) $$\begin{bmatrix} \dot{w} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} -2 & 197 \\ -0.25 & -15 \end{bmatrix} \begin{bmatrix} w \\ \theta \end{bmatrix} + \begin{bmatrix} -27 \\ -45 \end{bmatrix} [\delta]$$

(b) $\dot{x} = Ax + Bu$

Let $x = x^* e^{\lambda t}$ $\quad \therefore \dot{x} = \lambda x^* e^{\lambda t}$

$\therefore (A - \lambda I) x + Bu = 0$

$\therefore |A - \lambda I| = 0$

$$\therefore \begin{vmatrix} -2-\lambda & 197 \\ -0.25 & -15-\lambda \end{vmatrix} = \lambda^2 + 17\lambda + 79.25 = 0$$

$$\therefore \lambda = \frac{-17 \pm \sqrt{289 - 317}}{2} = -\frac{17}{2} \pm 2\sqrt{7}\, i$$

$\therefore$ Oscillatory stable

(c) $$\begin{bmatrix} \dot{\omega} \\ \ddot{\theta} \end{bmatrix} = \left(\begin{bmatrix} -2 & 197 \\ -0.25 & -15 \end{bmatrix} + \begin{bmatrix} -27 \\ -45 \end{bmatrix} \begin{bmatrix} K_1 & K_2 \end{bmatrix} \right) \begin{bmatrix} \omega \\ \theta \end{bmatrix}$$

$$= \begin{bmatrix} -2-27K_1 & 197-27K_2 \\ -0.25-45K_1 & -15-45K_2 \end{bmatrix} \begin{bmatrix} \omega \\ \theta \end{bmatrix}$$

$$\therefore \begin{vmatrix} -2-27K_1-\lambda & 197-27K_2 \\ -0.25-45K_1 & -15-45K_2-\lambda \end{vmatrix} = 0$$

$$\therefore [\lambda^2 + (27K_1 + 45K_2 + 17)\lambda + 1215K_1K_2 + 405K_1 + 90K_2 + 30]$$

$$-(1215K_1K_2 - 8865K_1 + 6.75K_2 - 49.25) = 0$$

$$\therefore \lambda^2 + (27K_1 + 45K_2 + 17)\lambda + 9270K_1 + 83.25K_2 + 79.25 = 0$$

$$\therefore \lambda = \frac{-(27K_1 + 45K_2 + 17) \pm \sqrt{(27K_1 + 45K_2 + 17)^2 - 4(9270K_1 + 83.25K_2 + 79.25)}}{2}$$

$$= \lambda \pm si$$

$$\therefore \zeta = \frac{-\lambda}{\omega_n} = \frac{-\lambda}{\sqrt{\lambda^2 + s^2}} = 0.5 \text{ while } \omega_n = 20 \text{ rad/s}$$

$$\because \lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0$$

$$\therefore \lambda^2 + 2\cdot 5\cdot 20\lambda + 20^2 = 0$$

$$\therefore \lambda^2 + 200\lambda + 40 = 0$$

$$\therefore \begin{cases} 27K_1 + 45K_2 + 17 = 200 \\ 9270K_1 + 83.25K_2 + 79.25 = 40 \end{cases}$$

$$\therefore \begin{cases} K_1 = 0.0342 \\ K_2 = 0.0462 \end{cases}$$

Review 1

5. Consider the pendulum on a cart system you analysed in tutorial sheet 1. The unstable equilibrium of this system can be controlled by applying an external axial force F_x in response to any change in angle θ about the desired equilibrium condition.

   (a) Express the equations governing the motion of the system in the state space form

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}.$$

   (b) For the mass and dimensions of the system presented in Tutorial 1, plot a root-locus plot to illustrate the effect of a proportional gain of the closed-loop dynamics of the system.

The equations of motion of a pendulum of mass m and length L, mounted on a cart of mass M, effectively a Segway, are

$$\ddot{\theta} = \frac{3}{2}\left(\frac{g\sin\theta - \ddot{x}\cos\theta}{L}\right)$$

$$\ddot{x}(M+m) + \frac{mL}{2}(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta) = 0$$

(a) $M\dot{x} = Ax + B\dot{x}$

$$\therefore \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & M+m \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$= \begin{bmatrix} \dot{\theta} \\ \frac{3}{2L}(g\sin\theta - \ddot{x}\cos\theta) \\ \frac{mL}{2}(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{2L}(g\cos\theta + \ddot{x}\sin\theta) & 0 & 0 \\ \frac{mL}{2}(-\ddot{\theta}\sin\theta - \dot{\theta}^2\cos\theta) & -mL\dot{\theta}\sin\theta & 0 \end{bmatrix}\begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -\frac{3}{2L}\cos\theta \\ -mL\dot{\theta}\sin\theta & \frac{mL}{2}\cos\theta & 0 \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & \frac{3}{2L}\cos\theta \\ mL\dot{\theta}\sin\theta & -\frac{mL}{2}\cos\theta & M+m \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{2L}(g\cos\theta + \ddot{x}\sin\theta) & 0 & 0 \\ \frac{mL}{2}(-\ddot{\theta}\sin\theta - \dot{\theta}^2\cos\theta) & -mL\dot{\theta}\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

For $M\dot{x} = Ax + Bu$,

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & \frac{3}{2L}\cos\theta \\ mL\dot{\theta}\sin\theta & -\frac{mL}{2}\cos\theta & M+m \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{2L}(g\cos\theta + \ddot{x}\sin\theta) & 0 & 0 \\ \frac{mL}{2}(-\ddot{\theta}\sin\theta - \dot{\theta}^2\cos\theta) & -mL\dot{\theta}\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

+ see solution

(b) see solution